high frequency products

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RF Amplifier ACOM 600S
Serial Port
Communication Protocol

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1. Structure of the messages between the amplifier and computer and protocol rules

Communication between the amplifier and computer is realized trough RS232 interface on the rear panel of the amplifier. Settings of RS232 interface are: baud rate 9600 8N1. Pause between messages is 10 ms. Max length of the messages is 72 bytes. When the computer receive message from the amplifier it have to return special confirmation message (message with address 0x86). If this message not received by the amplifier last message from the amplidier will be sent again. Each message initiated by the amplifier must be confirmed by the computer. Otherwise it will be sent again. If the message is not confirmed it will be discarded after the fifth attempt.

Structure of the communication protocol messages:

```
|Start TAG| Address | Length  |  - - - - D A T A - - - -  | CHKSUM  |
|0101 0101|xxxx xxxx|  N + 1  |xxxx xxxx|       |xxxx xxxx|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|       |---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | . . . | Byte N-1| Byte N  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – unique for each message
Byte 2 – Length of the message (all bytes including start byte ans check sum)
Byte 3..Byte N-1 – Data field
Byte N – Check Sum: CHKSUM

$Byte_{(N)} = 256 – \{SUM (Byte_{(0)} .. Byte_{(N-1)}) \& 0xFF\}$

If the message is received correctly $SUM (Byte_{(0)} .. Byte_{(N)}) = 0$.

2. Measurement units

Measurement units of parameters are noted in square brackets:

- [us] – microseconds
- [ms] – milliseconds
- [s] – seconds
- [mV] – millivolts
- [V] – volts
- [10 x V] – value is volts multiplied by 10. Example: 521 means U = 52.1 V
- [W] – watts
- [10 x W] – value is watts multiplied by 10. Example: 253 means U = 25.3 W
- [deg K] – temperature in degree Kelvins
- [deg C] – temperature in degree Celsius
- [deg F] – temperature in degree Fahreinheit
- [mA] – milliampers
- [A] – ampers
- [kHz] – kilohertz
- [100x] – value is the parameter multiplied by 100. Example: 217 means SWR = 2.17

3. Messages and commands from the Amplifier to the computer

3.1. Reply to the wrong received message:

Note: Generated after error in receiving!

Start TAG	Address	Length	CHKSUM
0101 0101	0000 0001	0000 0100	1010 0110
Byte 0	Byte 1	Byte 2	Byte 3

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x01
Byte 2 – Length of the message – 4 Bytes
Byte 3 – CHKSUM of the message – 0xA6

3.2. Request for specific message:

Start TAG	Address	Length	Data	CHKSUM
0101 0101	0000 0010	0000 0101	xxxx xxxx	cccc cccc
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x02
Byte 2 – Length of the message – 5 bytes
Byte 3 – Number of desired message
Byte 4 – CHKSUM of the message

3.3. System Configuration

3.3.1. PCB Versions of the main CPU module

Note: Generated once on request!

Start TAG	Address	Length	Ampl Type	FW Ver	FW SVer	CPU Ver
0101 0101	0001 0001	0001 1110	tttt tttt	vvvv vvvv	ssss ssss	xxxx xxxx
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

CPU SVer	IntA Ver	IntA SVer	IntB Ver	IntB SVer	Mod. Ver	Mod. SVer
xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx	xxxx xxxx
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13

BL Ver	BL SVer	A.SN [0]	A.SN [1]	A.SN [2]	A.SN [3]	A.SN [4]
xxxx xxxx	xxxx xxxx	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 14	Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20

A.SN [5]	A.SN [6]	A.SN [7]	A.SN [8]	A.SN [9]	A.SN [10]	A.SN [11]
ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 21	Byte 22	Byte 23	Byte 24	Byte 25	Byte 26	Byte 27

```
|HF Count | CHKSUM  |
|ffff ffff|cccc cccc|
|---- ----|---- ----|
| Byte 28 | Byte 29 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x11
Byte 2 – Length of the message – 30 bytes
Byte 3 – Amplifier Type (1 - A600S)
Byte 4 – Amplifier Firmware version
Byte 5 – Amplifier Frmware sub-version
Byte 6 – PCB CPU version
Byte 7 – PCB CPU sub-version
Byte 8 – PCB Interface_A version
Byte 9 – PCB Interface_A sub-version
Byte 10 – PCB Interface_B version
Byte 11 – PCB Interface_B sub-version
Byte 12 – PCB ATU Modem version (if the modem is mounted)
Byte 13 – PCB ATU Modem sub-version (if the modem is mounted)
Byte 14 –Bootloader version
Byte 15 –Bootloader sub-version
Byte 16..Byte 27 – Serial number of the amplifier
Byte 28 – Number of all Hard Fault records until now
Byte 29 – CHKSUM of the message

3.3.2. Settings of the Amplifier

Note: Generated once on request!

```
|Start TAG| Address | Length  |  CAT Settings 1   |  CAT Settings 2   |
|0101 0101|0001 0010|0000 1101|cccc cccc|cccc cccc|cccc cccc|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|  CAT Settings 3   |     User Flags    | Buz Vol | CHKSUM  |
|cccc cccc|cccc cccc|ffff ffff|ffff ffff|bbbb bbbb|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x12
Byte 2 – Length of the message – 13 bytes
Byte 3 – CAT Settings 1 (Low Byte)
    bit 4..bit 7 – CAT Baud Rate (valid only for CAT Interface with TTL Levels or RS232 Levels)
        0 – No CAT interface specified
        1 – 1200 bps
        2 – 4800 bps
        3 – 9600 bps
        4 – 19200 bps
        5 – 38400 bps
        6 – 57600 bps
    bit 0..bit 3 – Not Used
Byte 4 – CAT Settings 1 (High Byte)
    bit 4..bit 7 – CAT Interface Type (type of the CAT interface)
        0 – OFF

1 – RS232 Levels
2 – TTL Levels
3 – BCD Code
4 – RS232 Levels

bit 0..bit 3 – CAT Command Set (Commands Set)
0 – No CAT interface specified
1 – Commands Set 1
...
15 – Commands Set 15

Byte 5 – CAT Settings 2 (Low Byte)
Byte 6 – CAT Settings 2 (High Byte)
CAT Byte Spacing [us] (0..65535)
Byte 7 – CAT Settings 3 (Low Byte)
Byte 8 – CAT Settings 3 (High Byte)
CAT Polling Time [ms] (0..65535)
Byte 9 – User Flags (Low Byte):

bit 7 – (0x07)		
bit 6 – (0x06) Operate Access	0 – Disabled	1 – Enabled
bit 5 – (0x05) ON_BTN Pulses	0 – Long ON / Short OFF	1 – Short ON / Long OFF
bit 4 – (0x04) ATAC Procedure:	0 – Manual	1 – Auto
bit 3 – (0x03) CPU Buzzer:	0 – Disabled	1 – Enabled
bit 2 – (0x02) Temperature Flag:	0 – [º C]	1 – [º F]
bit 1 – (0x01) Auto Operate:	0 – Disabled	1 – Enabled
bit 0 – (0x00) Extra Cooling:	0 – Disabled	1 – Enabled

Byte 10 – User Flags (High Byte):
bit 7 – (0x0F) Last Power OFF is because of HARD FAULT Error
bit 6 – (0x0E)
bit 5 – (0x0D)
bit 4 – (0x0C)
bit 3 – (0x0B)
bit 2 – (0x0A)
bit 1 – (0x09)
bit 0 – (0x08)
Byte 11 – Buzzer Volume (0 – 100%)
Byte 12 – CHKSUM of the message

3.3.3. Serial number of the amplifier:

Note: Generated once on request!

Start TAG	Address	Length	A.SN [0]	A.SN [1]	A.SN [2]	A.SN [3]	A.SN [4]
0101 0101	0001 0011	0001 0000	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7

A.SN [5]	A.SN [6]	A.SN [7]	A.SN [8]	A.SN [9]	A.SN [10]	A.SN [11]	CHKSUM
ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	ssss ssss	cccc cccc
Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14	Byte 15

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x13
Byte 2 – Length of the message – 16 bytes
Byte 3..Byte 14 – Serial number of the amplifier
Byte 15 – CHKSUM of the message

3.3.4. User Sign, saved in EEPROM memory of the main CPU of the amplifier

Note: Generated once on request!

```
|Start TAG| Address | Length  |H.Msg[0] |H.Msg[1] |H.Msg[2] |H.Msg[3] |H.Msg[4] |
|0101 0101|0001 0100|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[5] |H.Msg[6] |H.Msg[7] |H.Msg[8] |H.Msg[9] |H.Msg[10]|H.Msg[11]|H.Msg[12]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[13]|H.Msg[14]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x14
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello Message – chars from 0 to 14
Byte 15 – CHKSUM of the message

```
|Start TAG| Address | Length  |H.Msg[15]|H.Msg[16]|H.Msg[17]|H.Msg[18]|H.Msg[19]|
|0101 0101|0001 0101|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[20]|H.Msg[21]|H.Msg[22]|H.Msg[23]|H.Msg[24]|H.Msg[25]|H.Msg[26]|H.Msg[27]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[28]|H.Msg[29]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x15
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 14 – Hello message – signs from 15 to 29
Byte 15 – CHKSUM of the message

3.3.5. ATU (Antenna Tuner) Parameters:

Note: Generated once on request!

```
|Start TAG| Address | Length  |ATU Type | HW Ver  | HW SVer | FW Ver  | FW SVer |
|0101 0101|0001 0110|0001 0011|tttt tttt|tttt tttt|ssss ssss|vvvv vvvv|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|ATUSN[0] |ATUSN[1] |ATUSN[2] |ATUSN[3] |ATUSN[4] |ATUSN[5] |ATUSN[6] |ATUSN[7] |
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |
```

```
|ATUSN[8] |ATUSN[9] | CHKSUM  |
|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x16
Byte 2 – Length of the message – 19 bytes
Byte 3 – ATU Type
Byte 4 – ATU Hardware version
Byte 5 – ATU Hardware sub-version
Byte 6 – ATU firmware version
Byte 7 – ATU firmware sub-version
Byte 8, Byte 17 – Serial number of the ATU
Byte 18 – CHKSUM of the message

3.3.6. ASEL (Antenna Selector) Parameters:

Note: Generated once on request!

```
|Start TAG| Address | Length  |ASEL Type| HW Ver  | HW SVer | FW Ver  | FW SVer |
|0101 0101|0001 0111|0001 0011|tttt tttt|tttt tttt|ssss ssss|vvvv vvvv|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|ASELSN[0]|ASELSN[1]|ASELSN[2]|ASELSN[3]|ASELSN[4]|ASELSN[5]|ASELSN[6]|ASELSN[7]|
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|ASELSN[8]|ASELSN[9]| CHKSUM  |
|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x17
Byte 2 – Length of the message – 19 bytes
Byte 3 – ASEL Type
Byte 4 – ASEL hardware version
Byte 5 – ASEL hardware sub-version
Byte 6 – ASEL firmware version
Byte 7 – ASEL firmware sub-version
Byte 8, Byte 17 – Serial number of the ASEL
Byte 18 – CHKSUM of the message

3.3.7. Antenna names saved in EEPROM memory in main CPU

Note: Generated once on request!

```
|Start TAG| Address | Length  |Name [0] |Name [1] |Name [2] |Name [3] |Name [4] |
|0101 0101|0001 1000|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |
```

```
|Name [5] |Name [6] |Name [7] |Name [8] |80m/160m |40m/60m  |20m/30m  |15m/17m  |
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|bbbb BBBB|bbbb BBBB|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|10m/12m  |Spare/6m | CHKSUM  |
|bbbb BBBB|bbbb BBBB|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x18
Byte 2 – Length of the message – 19 bytes
Byte 3 ÷ Byte 11 – Antenna Name
Byte 12 – Antenna Assigment and Selectivity for Bands 80m and 160m (3,5 MHz and 1,8 MHz)
- bit 0 - bit 2 - Antenna Type for 160m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 160m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 6 - Antenna Type for 80m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 7 - Antenna Assigment for 80m [0 – Not Assigned, 1 – Assigned]

Byte 13 – Antenna Assigment and Selectivity for Bands 40m and 60m (7,0 MHz and 5,0 MHz)
- bit 0 - bit 2 - Antenna Type for 60m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 60m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 6 - Antenna Type for 40m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 7 - Antenna Assigment for 40m [0 – Not Assigned, 1 – Assigned]

Byte 14 – Antenna Assigment and Selectivity for Bands 20m and 30m (14 MHz and 10 MHz)
- bit 0 - bit 2 - Antenna Type for 30m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 30m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 6 - Antenna Type for 20m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 7 - Antenna Assigment for 20m [0 – Not Assigned, 1 – Assigned]

Byte 15 – Antenna Assigment and Selectivity for Bands 15m and 17m (21 MHz and 18 MHz)
- bit 0 - bit 2 - Antenna Type for 17m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 17m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 6 - Antenna Type for 15m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 7 - Antenna Assigment for 15m [0 – Not Assigned, 1 – Assigned]

Byte 16 – Antenna Assigment and Selectivity for Bands 10m and 12m (28 MHz and 24 MHz)
- bit 0 - bit 2 - Antenna Type for 12m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 12m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 6 - Antenna Type for 10m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 7 - Antenna Assigment for 10m [0 – Not Assigned, 1 – Assigned]

Byte 17 – Antenna Assigment and Selectivity for Band 6m (50 MHz)
- bit 0 - bit 2 - Antenna Type for 6m [0 - Narrow, 1 - Default, 2 - Wide, 3 - Exact]
- bit 3 - Antenna Assigment for 6m [0 – Not Assigned, 1 – Assigned]
- bit 4 - bit 7 - Not Used

Byte 18 – CHKSUM of the message

3.3.8. Fixed values of bias voltages of the transistors (Obsolete! Removed)

Note: Generated once on request!

```
|Start TAG| Address | Length  |  BIAS_1A_SET [mV] |  BIAS_1B_SET [mV] |
|0101 0101|0001 1001|0001 0100|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|  BIAS_1C_SET [mV] |  BIAS_1D_SET [mV] |  BIAS_2A_SET [mV] |  BIAS_2B_SET [mV] |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
```

```
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|  BIAS_2C_SET [mV] |  BIAS_2D_SET [mV] | CHKSUM  |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x19
Byte 2 – Length of the message – 20 bytes
Byte 3 – BIAS_1A_SET [mV] (Low Byte) (Not Used)
Byte 4 – BIAS_1A_SET [mV] (High Byte) (Not Used)
Byte 5 – BIAS_1B_SET [mV] (Low Byte) (Not Used)
Byte 6 – BIAS_1B_SET [mV] (High Byte) (Not Used)
Byte 7 – BIAS_1C_SET [mV] (Low Byte) (Not Used)
Byte 8 – BIAS_1C_SET [mV] (High Byte) (Not Used)
Byte 9 – BIAS_1D_SET [mV] (Low Byte) (Not Used)
Byte 10 – BIAS_1D_SET [mV] (High Byte) (Not Used)
Byte 11 – BIAS_2A_SET [mV] (Low Byte) (Not Used)
Byte 12 – BIAS_2A_SET [mV] (Hugh Byte) (Not Used)
Byte 13 – BIAS_2B_SET [mV] (Low Byte) (Not Used)
Byte 14 – BIAS_2B_SET [mV] (High Byte) (Not Used)
Byte 15 – BIAS_2C_SET [mV] (Low Byte) (Not Used)
Byte 16 – BIAS_2C_SET [mV] (High Byte) (Not Used)
Byte 17 – BIAS_2D_SET [mV] (Low Byte) (Not Used)
Byte 18 – BIAS_2D_SET [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.4. Parameters of the amplifier:

3.4.1. Error Codes (ErrorCodes0 ... ErrorCodes9):

Note: Generated once when new alarm was appeared or at any time on request!

```
|Start TAG| Address | Length  |   Error Codes 0   |   Error Codes 1   |
|0101 0101|0010 0001|0001 1000|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|   Error Codes 2   |   Error Codes 3   |   Error Codes 4   |   Error Codes 5   |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|   Error Codes 6   |   Error Codes 7   |   Error Codes 8   |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |

|   Error Codes 9   | CHKSUM  |
|eeee eeee|eeee eeee|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 21 | Byte 22 | Byte 23 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x21
Byte 2 – Length of the message – 24 bytes

Byte 3 – Error Codes 0 (Low Byte):

bit 7 - (0x07) EXCESSIVE DRIVE POWER	(SOFT FAULT)
bit 6 - (0x06) DRIVE POWER TOO HIGH	(WARNING)
bit 5 - (0x05) EXCESSIVE REFLECTED POWER	(SOFT FAULT)
bit 4 - (0x04) REFLECTED POWER WARNING	(WARNING)
bit 3 - (0x03) DRIVE POWER DETECTED AT WRONG TIME	(HARD FAULT)
bit 2 - (0x02) OUTPUT RELAY OPEN IT SHOULD BE CLOSED	(HARD FAULT)
bit 1 - (0x01) OUTPUT RELAY CLOSED IT SHOULD BE OPEN	(WARNING)
bit 0 - (0x00) HOT SWITCHING ATTEMPT	(WARNING)

Byte 4 – Error Codes 0 (High Byte):

bit 7 - (0x0F) REMOVE DRIVE POWER IMMEDIATELY	(WARNING)
bit 6 - (0x0E) STOP TRANSMISSION FIRST	(WARNING)
bit 5 - (0x0D) PA LOAD SWR TOO HIGH	(SOFT FAULT)
bit 4 - (0x0C) DETECTED RF POWER AT WRONG TIME	(WARNING)
bit 3 - (0x0B) OUTPUT DISBALANCE	(SOFT FAULT)
bit 2 - (0x0A) FREQUENCY VIOLATION	(SOFT FAULT)
bit 1 - (0x09) DRIVE FREQUENCY OUT OF RANGE	(SOFT FAULT)
bit 0 - (0x08) HOT SWITCHING ATTEMPT	(SOFT FAULT)

Byte 5 – Error Codes 1 (Low Byte):

bit 7 - (0x17) LPF FAN SPEED TOO LOW	(HARD FAULT)
bit 6 - (0x16) PAM2 FAN SPEED TOO LOW	(HARD FAULT)
bit 5 - (0x15) PAM1 FAN SPEED TOO LOW	(HARD FAULT)
bit 4 - (0x14)	
bit 3 - (0x13) 26V TOO HIGH	(HARD FAULT)
bit 2 - (0x12) 26V TOO LOW	(HARD FAULT)
bit 1 - (0x11) 5V TOO HIGH	(HARD FAULT)
bit 0 - (0x10) 5V TOO LOW	(HARD FAULT)

Byte 6 – Error Codes 1 (High Byte):

bit 7 - (0x1F) PAM2 EXCESSIVE TEMPERATURE	(SOFT FAULT)
bit 6 - (0x1E) PAM1 EXCESSIVE TEMPERATURE	(SOFT FAULT)
bit 5 - (0x1D) PAM2 TEMPERATURE TOO HIGH	(WARNING)
bit 4 - (0x1C) PAM1 TEMPERATURE TOO HIGH	(WARNING)
bit 3 - (0x1B) PAM2 DISSIPATION POWER WARNING	(WARNING)
bit 2 - (0x1A) PAM1 DISSIPATION POWER WARNING	(WARNING)
bit 1 - (0x19) PAM2 DISSIPATION POWER TOO HIGH	(SOFT FAULT)
bit 0 - (0x18) PAM1 DISSIPATION POWER TOO HIGH	(SOFT FAULT)

Byte 7 – Error Codes 2 (Low Byte):

bit 7 - (0x27) BIAS_1B VOLTAGE ERROR	(SOFT FAULT)
bit 6 - (0x26) BIAS_1A VOLTAGE ERROR	(SOFT FAULT)
bit 5 - (0x25) PAM1 EXCESSIVE CURRENT	(SOFT FAULT)
bit 4 - (0x24) PAM1 CURRENT WARNING	(WARNING)
bit 3 - (0x23) PAM1 IDLE CURRENT TOO LOW	(SOFT FAULT)
bit 2 - (0x22) PAM1 CURRENT = …mA SHOULD BE ZERO	(SOFT FAULT)
bit 1 - (0x21) PAM1 HV TOO HIGH	(HARD FAULT)
bit 0 - (0x20) PAM1 HV TOO LOW	(HARD FAULT)

Byte 8 – Error Codes 2 (High Byte):

bit 7 - (0x2F) PAM1 GAIN TOO HIGH	(SOFT FAULT)
bit 6 - (0x2E) PAM1 GAIN TOO LOW	(SOFT FAULT)
bit 5 - (0x2D) BIAS_1D = …V SHOULD BE ZERO	(HARD FAULT)
bit 4 - (0x2C) BIAS_1C = …V SHOULD BE ZERO	(HARD FAULT)
bit 3 - (0x2B) BIAS_1B = …V SHOULD BE ZERO	(HARD FAULT)
bit 2 - (0x2A) BIAS_1A = …V SHOULD BE ZERO	(HARD FAULT)
bit 1 - (0x29) BIAS_1D VOLTAGE ERROR	(SOFT FAULT)
bit 0 - (0x28) BIAS_1C VOLTAGE ERROR	(SOFT FAULT)

Byte 9 – Error Codes 3 (Low Byte):

bit 7 - (0x37) BIAS_1D = …V SHOULD BE ZERO (HARD FAULT)
bit 6 - (0x36) BIAS_1C = …V SHOULD BE ZERO (HARD FAULT)
bit 5 - (0x35) BIAS_1B = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x34) BIAS_1A = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x33) PAM1 TEMPERATURE TOO HIGH (WARNING)
bit 2 - (0x32) PAM1 EXCESSIVE TEMPERATURE (HARD FAULT)
bit 1 - (0x31) PAM1 CURRENT = …mA SHOULD BE ZERO (HARD FAULT)
bit 0 - (0x30) PAM1 HV = …V SHOULD BE ZERO (HARD FAULT)

Byte 10 – Error Codes 3 (High Byte):

bit 7 - (0x3F)
bit 6 - (0x3E)
bit 5 - (0x3D)
bit 4 - (0x3C)
bit 3 - (0x3B)
bit 2 - (0x3A)
bit 1 - (0x39) PAM1 EXCESSIVE CURRENT. CHECK SWR OR REDUCE DRIVE POWER (SOFT FAULT)
bit 0 - (0x38) PSU1 EXCESSIVE TEMPERATURE (WARNING)

Byte 11 – Error Codes 4 (Low Byte):

bit 7 - (0x47) BIAS_2B VOLTAGE ERROR (SOFT FAULT)
bit 6 - (0x46) BIAS_2A VOLTAGE ERROR (SOFT FAULT)
bit 5 - (0x45) PAM2 EXCESSIVE CURRENT (SOFT FAULT)
bit 4 - (0x44) PAM2 CURRENT WARNING (WARNING)
bit 3 - (0x43) PAM2 IDLE CURRENT TOO LOW (SOFT FAULT)
bit 2 - (0x42) PAM2 CURRENT = …mA SHOULD BE ZERO (SOFT FAULT)
bit 1 - (0x41) PAM2 HV TOO HIGH (HARD FAULT)
bit 0 - (0x40) PAM2 HV TOO LOW (HARD FAULT)

Byte 12 – Error Codes 4 (High Byte):

bit 7 - (0x4F) PAM2 GAIN TOO HIGH (SOFT FAULT)
bit 6 - (0x4E) PAM2 GAIN TOO LOW (SOFT FAULT)
bit 5 - (0x4D) BIAS_2D = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x4C) BIAS_2C = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x4B) BIAS_2B = …V SHOULD BE ZERO (HARD FAULT)
bit 2 - (0x4A) BIAS_2A = …V SHOULD BE ZERO (HARD FAULT)
bit 1 - (0x49) BIAS_2D VOLTAGE ERROR (SOFT FAULT)
bit 0 - (0x48) BIAS_2C VOLTAGE ERROR (SOFT FAULT)

Byte 13 – Error Codes 5 (Low Byte):

bit 7 - (0x57) BIAS_2D = …V SHOULD BE ZERO (HARD FAULT)
bit 6 - (0x56) BIAS_2C = …V SHOULD BE ZERO (HARD FAULT)
bit 5 - (0x55) BIAS_2B = …V SHOULD BE ZERO (HARD FAULT)
bit 4 - (0x54) BIAS_2A = …V SHOULD BE ZERO (HARD FAULT)
bit 3 - (0x53) PAM2 TEMPERATURE TOO HIGH (WARNING)
bit 2 - (0x52) PAM2 EXCESSIVE TEMPERATURE (HARD FAULT)
bit 1 - (0x51) PAM2 CURRENT = …mA SHOULD BE ZERO (HARD FAULT)
bit 0 - (0x50) PAM2 HV = …V SHOULD BE ZERO (HARD FAULT)

Byte 14 – Error Codes 5 (High Byte):

bit 7 - (0x3F)
bit 6 - (0x5E)
bit 5 - (0x5D)
bit 4 - (0x5C)
bit 3 - (0x5B)
bit 2 - (0x5A)
bit 1 - (0x59) PAM2 EXCESSIVE CURRENT. CHECK SWR OR REDUCE DRIVE POWER

(SOFT FAULT)
bit 0 - (0x58) PSU2 EXCESSIVE TEMPERTURE (WARNING)

Byte 15 – Error Codes 6 (Low Byte):
bit 7 - (0x67) ATU POWER SWITCH ALARM AT POWER ON (WARNING)
bit 6 - (0x66) ATU POWER SWITCH ALARM (WARNING)
bit 5 - (0x65) ATU MODEM EXCESSIVE TEMPERATURE (WARNING)
bit 4 - (0x64) DISPLAY UNIT COMMUNICATION ERROR (WARNING)
bit 3 - (0x63) PSU2 EXCESSIVE TEMPERATURE (SOFT FAULT)
bit 2 - (0x62) PSU1 EXCESSIVE TEMPERATURE (SOFT FAULT)
bit 1 - (0x61) PSU2/CONTROL MALFUNCTION (HARD FAULT)
bit 0 - (0x60) PSU1/CONTROL MALFUNCTION (HARD FAULT)

Byte 16 – Error Codes 6 (High Byte):
bit 7 - (0x6F) LOSS OF EEPROM DATA (SOFT FAULT)
bit 6 - (0x6E) LOSS OF EEPROM DATA (WARNING)
bit 5 - (0x6D) LOSS OF ETHERNET SETTINGS (WARNING)
bit 4 - (0x6C) LOSS OF AUDIO DATA (WARNING)
bit 3 - (0x6B)
bit 2 - (0x6A)
bit 1 - (0x69) AUDIO MEMORY NOT RESPONDING (WARNING)
bit 0 - (0x68) ETHERNET CONTROLLER NOT RESPONDING (WARNING)

Byte 17 – Error Codes 7 (Low Byte):
bit 7 - (0x77)
bit 6 - (0x76)
bit 5 - (0x75)
bit 4 - (0x74)
bit 3 - (0x73)
bit 2 - (0x72)
bit 1 - (0x71)
bit 0 - (0x70) CAT ERROR (WARNING)

Byte 18 – Error Codes 7 (High Byte):
bit 7 - (0x7F)
bit 6 - (0x7E)
bit 5 - (0x7D)
bit 4 - (0x7C)
bit 3 - (0x7B)
bit 2 - (0x7A)
bit 1 - (0x79)
bit 0 - (0x78)

Byte 19 – Error Codes 8 (Low Byte):
bit 7 - (0x87) NO ANTENNA SETTINGS PREPARED (WARNING)
bit 6 - (0x86) NO TUNUNG SETTINGS PREPARED (WARNING)
bit 5 - (0x85) AMP-ASEL COMMUNICATION ERROR (WARNING)
bit 4 - (0x84) ASEL-AMP COMMUNICATION ERROR (WARNING)
bit 3 - (0x83) ASEL NOT RESPONDING (WARNING)
bit 2 - (0x82) AMP-ATU COMMUNICATION ERROR (WARNING)
bit 1 - (0x81) ATU-AMP COMMUNICATION ERROR (WARNING)
bit 0 - (0x80) ATU NOT RESPONDING (WARNING)

Byte 20 – Error Codes 8 (High Byte):
bit 7 - (0x8F)
bit 6 - (0x8E)
bit 5 - (0x8D)
bit 4 - (0x8C)
bit 3 - (0x8B) ATU MEMORY FAIL (WARNING)
bit 2 - (0x8A) ATU TUNING CYCLE UNSUCCESSFUL (WARNING)

bit 1 - (0x89) ANTENNA CAN NOT BE CHANGED UNTIL RF POWER IS PRESENT (WARNING)
bit 0 - (0x88) ATU CAN NOT BE RE-TUNED UNTIL RF POWER IS PRESENT

Byte 21 – Error Codes 9 (Low Byte):

bit 7 - (0x87) BIAS_2D = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 6 - (0x86) BIAS_2C = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 5 - (0x85) BIAS_2B = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 4 - (0x84) BIAS_2A = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 3 - (0x83) BIAS_1D = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 2 - (0x82) BIAS_1C = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 1 - (0x81) BIAS_1B = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)
bit 0 - (0x80) BIAS_1A = …V SHOULD BE BETWEEN L_TH V AND H_TH V. IF OK SELECT OFF (WARNING)

Byte 22 – Error Codes 9 (High Byte):

bit 7 - (0x8F) BIAS_2D = …V SHOULD BE BELOW H_TH V (WARNING)
bit 6 - (0x8E) BIAS_2C = …V SHOULD BE BELOW H_TH V (WARNING)
bit 5 - (0x8D) BIAS_2B = …V SHOULD BE BELOW H_TH V (WARNING)
bit 4 - (0x8C) BIAS_2A = …V SHOULD BE BELOW H_TH V (WARNING)
bit 3 - (0x8B) BIAS_1D = …V SHOULD BE BELOW H_TH V (WARNING)
bit 2 - (0x8A) BIAS_1C = …V SHOULD BE BELOW H_TH V (WARNING)
bit 1 - (0x89) BIAS_1B = …V SHOULD BE BELOW H_TH V (WARNING)
bit 0 - (0x88) BIAS_1A = …V SHOULD BE BELOW H_TH V (WARNING)

Byte 23 – CHKSUM of the message

3.4.2. Parametersfo the amplifier (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |Amp_State| Amplifier Flags 1 | Amplifier Flags 2 |
|0101 0101|0010 0011|0001 0100|ssss ssss|ffff ffff|ffff ffff|ffff ffff|ffff ffff|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|  P_DC 1 [10 x W]  |  P_DC 2 [10 x W]  |System Clock-HW [s]|System Clock-LW [s]|
|PPPP PPPP|PPPP PPPP|pppp pppp|pppp pppp|hhhh hhhh|hhhh hhhh|llll llll|llll llll|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|   temp1  [deg K]  |   temp2  [deg K]  | CHKSUM  |
|tttt tttt|tttt tttt|tttt tttt|tttt tttt|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x23
Byte 2 – Length of the message – 21 bytes
Byte 3 – Amplifier's mode

0x1X – Amplifier Reset
0x2X – Init Mode
0x3X – Debug Mode
0x4X – Service Mode
0x5X – STB Mode
0x6X – OPR/RX Mode
0x7X – OPR/TX Mode
0x8X – ATAC (ATU Tune Antenna Change) Procedure
0x9X – Set System Parameters Menu
0xAX – Turn OFF Procedure
0xA2 – Extra Cooling State

Byte 4 – Amplifier Flags 1 (Low Byte)
- bit 7 – FLAG_PwrOff_Request — Final Request for Power OFF
- bit 6 – FLAG_Soft_PwrOff — Request for Power OFF by Software
- bit 5 – FLAG_PWRON_RS — Amplifier is Turned ON by RS232
- bit 4 – FLAG_PWRON_RMT — Amplifier is Turned ON by Remote Voltage
- bit 3 – FLAG_PWRON_BTN — Amplifier is Turned ON by Panel Button
- bit 2 – FLAG_ATAC_InProgress — ATAC Procedure is in Progress
- bit 1 – FLAG_F_RANGE_ERR — Input Frequency is Out of Range (from F-Counter)
- bit 0 – FLAG_LOW_IN_PWR — Low Input Power (from F-Counter)

Byte 5 – Amplifier Flags 1 (High Byte)
- bit 15 – FLAG_Last_Com — Last Command Source (0 - Local / 1 - Remote)
- bit 14 – FLAG_ATAC_REQ — Request for ATAC Procedure
- bit 13 – FLAG_HV_ON — Status of High Voltage Control Signal
- bit 12 – FLAG_HV_MON — Enable/Disable High Voltage Monitoring
- bit 11 – FLAG_HV_CTRL — Request for High Voltage ON/OFF Procedures
- bit 10 – FLAG_BIASEN — Status of Bias Control Signal
- bit 9 – FLAG_BIAS_MON — Enable/Disable Bias Voltages Monitoring
- bit 8 – FLAG_BIAS_CTRL — Request for Bias Voltages ON/OFF Procedures

Byte 6 – Amplifier Flags 2 (Low Byte)
- bit 7 – FLAG_ORC — ORC Condition After ORC Detection Procedure
- bit 6 – FLAG_ORC_MON — Enable/Disable ORC Monitoring
- bit 5 – FLAG_OUTR — Status of the Output Relay Control Signal
- bit 4 – FLAG_OUTR_CTRL — Request for Output Relay ON/OFF Procedures
- bit 3 – FLAG_INR — Status of the Input Relay Control Signal
- bit 2 – FLAG_INR_CTRL — Request for Input Relay ON/OFF Procedures
- bit 1 – FLAG_KEYIN — Detected Level of the KEY IN Signal
- bit 0 – FLAG_KEYIN_MON — Enable/Disable KEY IN Monitoring

Byte 7 – Amplifier Flags 2 (High Byte)
- bit 15 – FLAG_WRN_DIS — Disable Warning Messages
- bit 14 – FLAG_SF_ERR_DIS — Disable SOFT FAULT Errors
- bit 13 – FLAG_HF_ERR_DIS — Disable HARD FAULT Errors
- bit 12 – Not Used
- bit 11 – FLAG_I_LKG_MON — ON/OFF Alarms Check During Input Relay Switching
- bit 10 – FLAG_ASEL_TUN — Indicate Whether ASEL is Tuned or Not (if associated)
- bit 9 – FLAG_ATU_TUN — Indicate Whether ATU is Tuned or Not (if associated)
- bit 8 – FLAG_LPF_TUN — Indicate Whether LPFs are Tuned or Not

Byte 8 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 9 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 10 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 11 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 12 – System Clock – High Word [sec] (Low Byte)
Byte 13 – System Clock – High Word [sec] (High Byte)
Byte 14 – System Clock – Low Word [sec] (Low Byte)

Byte 15 – System Clock – Low Word [sec] (High Byte)
Byte 16 – Temperature of PAM1 [deg K] (Low Byte)
Byte 17 – Temperature of PAM1 [deg K] (High Byte)
Byte 18 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 19 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 20 – CHKSUM of the message

3.4.3. Input power, forward power and reflected power of the amplifier (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |Input Power [10xW] | Forward Power [W] |
|0101 0101|0010 0100|0001 0100|iiii iiii|iiii iiii|oooo oooo|oooo oooo|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|Reflected Power [W]|     SWR [100x]    |PAM1 Dis.Pwr.[10xW]|PAM2 Dis.Pwr.[10xW]|
|rrrr rrrr|rrrr rrrr|ssss ssss|ssss ssss|pppp pppp|pppp pppp|pppp pppp|pppp pppp|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|    bdata  [mV]    |  disbalance [mV]  |  CHKSUM |
|bbbb bbbb|bbbb bbbb|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x24
Byte 2 – Length of the message – 20 bytes
Byte 3 – Input Power [10 x W] (Low Byte)
Byte 4 – Input Power [10 x W] (High Byte)
Byte 5 – Forward Power [W] (Low Byte)
Byte 6 – Forward Power [W] (High Byte)
Byte 7 – Reflected Power [W] (Low Byte)
Byte 8 – Reflected Power [W] (High Byte)
Byte 9 – SWR [100x] (Low Byte)
Byte 10 – SWR [100x] (High Byte)
Byte 11 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte)
Byte 12 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte)
Byte 13 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte) (Not Used)
Byte 14 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte) (Not Used)
Byte 15 – bdata [mV] (Low Byte)
Byte 16 – bdata [mV] (High Byte)
Byte 17 – disbalance [mV] (Low Byte) (Not Used)
Byte 18 – disbalance [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.4.4. Power supply voltages, currents and carrier frequency (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |      VCC5 [mV]    |     VCC26 [mV]    |
|0101 0101|0010 0101|0001 0101|vvvv vvvv|vvvv vvvv|vvvv vvvv|vvvv vvvv|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|    HV1 [10 x V]   |    HV2 [10 x V]   |     ID1 [mA]      |     ID2 [mA]      |
|HHHH HHHH|HHHH HHHH|hhhh hhhh|hhhh hhhh|iiii iiii|iiii iiii|iiii iiii|iiii iiii|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|  Frequency [kHz]  |ErrorCode|ErrPrm.LW|ErrPrm.HW| CHKSUM  |
|ffff ffff|ffff ffff|eeee eeee|pppp pppp|pppp pppp|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x25
Byte 2 – Length of the message – 21 bytes
Byte 3 – VCC5 [mV] (Low Byte)
Byte 4 – VCC5 [mV] (High Byte)
Byte 5 – VCC26 [10 x V] (Low Byte)
Byte 6 – VCC26 [10 x V] (High Byte)
Byte 7 – HV1 [10 x V] (Low Byte)
Byte 8 – HV1 [10 x V] (High Byte)
Byte 9 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 10 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 11 – ID1 [mA] (Low Byte)
Byte 12 – ID1 [mA] (High Byte)
Byte 13 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 14 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 15 – Carrier frequency [kHz] (Low Byte)
Byte 16 – Carrier frequency [kHz] (High Byte)
Byte 17 – Error code of the alarm condition displayed on the amplifier's screen
Byte 18 – Parameter of the alarm condition (Low Byte)
Byte 19 – Parameter of the alarm condition (High Byte)
Byte 20 – CHKSUM of the message

3.4.5. Bias voltages (Obsolete! Replaced by message 0x2F)

Note: Messages with addresses 0x23, 0x24, 0x25 and 0x26 are transmitted consecutive with 10 ms pause between messages!

```
|Start TAG| Address | Length  |    bias_1a [mV]   |    bias_1b [mV]   |
|0101 0101|0010 0110|0001 0000|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|    bias_1c [mV]   |    bias_1d [mV]   |    bias_2a [mV]   |    bias_2b [mV]   |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

|    bias_2c [mV]   |    bias_2d [mV]   | CHKSUM  |
|cccc cccc|cccc cccc|dddd dddd|dddd dddd|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 |
```

Byte 0 – Start byte– 0x55

Byte 1 – Address – 0x26
Byte 2 – Length of the message – 20 bytes
Byte 3 – bias_1a [mV] (Low Byte) (A600S)
Byte 4 – bias_1a [mV] (High Byte) (A600S)
Byte 5 – bias_1b [mV] (Low Byte) (A600S)
Byte 6 – bias_1b [mV] (High Byte) (A600S)
Byte 7 – bias_1c [mV] (Low Byte) (Not Used)
Byte 8 – bias_1c [mV] (High Byte) (Not Used)
Byte 9 – bias_1d [mV] (Low Byte) (Not Used)
Byte 10 – bias_1d [mV] (High Byte) (Not Used)
Byte 11 – bias_2a [mV] (Low Byte) (Not Used)
Byte 12 – bias_2a [mV] (High Byte) (Not Used)
Byte 13 – bias_2b [mV] (Low Byte) (Not Used)
Byte 14 – bias_2b [mV] (High Byte) (Not Used)
Byte 15 – bias_2c [mV] (Low Byte) (Not Used)
Byte 16 – bias_2c [mV] (High Byte) (Not Used)
Byte 17 – bias_2d [mV] (Low Byte) (Not Used)
Byte 18 – bias_2d [mV] (High Byte) (Not Used)
Byte 19 – CHKSUM of the message

3.4.6. Antenna number, selected band and band borders

Note: Generated only when band or band segmend was changed!

```
|Start TAG| Address | Length  | Ant Num |Ant Type |  Band   |Segm Cnt |
|0101 0101|0010 0111|0000 1101|nnnn nnnn|tttt tttt|bbbb bbbb|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

|Segm Num | Displ Band L_Freq | Displ Band H_Freq | Displ Segm L_Freq |
|nnnn nnnn|llll llll|llll llll|hhhh hhhh|hhhh hhhh|llll llll|llll llll|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 |

| Displ Segm H_Freq | CHKSUM  |
|hhhh hhhh|hhhh hhhh|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 14 | Byte 15 | Byte 16 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x27
Byte 2 – Length of the message – 17 bytes
Byte 3 – Number of the selected antenna [1..10]
Byte 4 – Type of the selected abtenna [1 - Default, 2 - Exact, 3 - Narrow, 4 - Wide]
Byte 5 – Number of the selected band
Byte 6 – Number of segments in selected band (from 1 to 80)
Byte 7 – Number of the current segment
Byte 8 – Low band border for visualization [kHz] (Low Byte)
Byte 9 – Low band border for visualization [kHz] (High Byte)
Byte 10 – High band border for visualization [kHz] (Low Byte)

Byte 11 – High band border for visualization [kHz] (High Byte)
Byte 12 – Low band border for visualization [kHz] (Low Byte)
Byte 13 – Low band border for visualization [kHz] (High Byte)
Byte 14 – High band border for visualization [kHz] (Low Byte)
Byte 15 – High band border for visualization [kHz] (High Byte)
Byte 16 – CHKSUM of the message

Note: Band borders are different, depends of whether amplifier is locked or not.

3.4.7. Band segments condition

Note: Generated once when band or band segmend was changed!

```
|Start TAG| Address | Length  |Seg Stat |Seg Stat |Seg Stat |Seg Stat |Seg Stat |
|0101 0101|0010 1000|0000 1110|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|Seg Stat |Seg Stat |Seg Stat |Seg Stat |Seg Stat | CHKSUM  |
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x28
Byte 2 – Length of the message – 14 bytes
Byte 3 – Byte 12 – band segments condition (0 – not tuned , 1 – tuned)
Byte 13 – CHKSUM of the message

3.4.8. Parameter of the amplifier (united messages: 0x23, 0x24, 0x25 and 0x26)

Note: Generated automaticly with 10 ms pause between messages!

```
|Start TAG| Address | Length  |Amp_State| Amplifier Flags 1 | Amplifier Flags 2 |
|0101 0101|0010 1111|0100 1000|ssss ssss|ffff ffff|ffff ffff|ffff ffff|ffff ffff|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|  P_DC 1 [10 x W]  |  P_DC 2 [10 x W]  |System Clock-HW [s]|System Clock-LW [s]|
|PPPP PPPP|PPPP PPPP|pppp pppp|pppp pppp|hhhh hhhh|hhhh hhhh|llll llll|llll llll|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|   temp1  [deg K]  |   temp2  [deg K]  |Input Power [10x W]| Forward Power [W] |
|tttt tttt|tttt tttt|tttt tttt|tttt tttt|iiii iiii|iiii iiii|oooo oooo|oooo oooo|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 | Byte 23 |

|Reflected Power [W]|     SWR [100x]    |PAM1 Dis.Pwr.[10xW]|PAM2 Dis.Pwr.[10xW]|
|rrrr rrrr|rrrr rrrr|ssss ssss|ssss ssss|pppp pppp|pppp pppp|pppp pppp|pppp pppp|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 24 | Byte 25 | Byte 26 | Byte 27 | Byte 28 | Byte 29 | Byte 30 | Byte 31 |

|    bdata  [mV]    |  disbalance [mV]  |      VCC5 [mV]    |     VCC26 [mV]    |
|bbbb bbbb|bbbb bbbb|dddd dddd|dddd dddd|vvvv vvvv|vvvv vvvv|vvvv vvvv|vvvv vvvv|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 32 | Byte 33 | Byte 34 | Byte 35 | Byte 36 | Byte 37 | Byte 38 | Byte 39 |

|    HV1 [10 x V]   |    HV2 [10 x V]   |     ID1 [mA]      |     ID2 [mA]      |
```

```
|HHHH HHHH|HHHH HHHH|hhhh hhhh|hhhh hhhh|iiii iiii|iiii iiii|iiii iiii|iiii iiii|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 40 | Byte 41 | Byte 42 | Byte 43 | Byte 44 | Byte 45 | Byte 46 | Byte 47 |

|  Frequency [kHz]  |    bias_1a [mV]   |    bias_1b [mV]   |    bias_1c [mV]   |
|ffff ffff|ffff ffff|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|cccc cccc|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 48 | Byte 49 | Byte 50 | Byte 51 | Byte 52 | Byte 53 | Byte 54 | Byte 55 |

|    bias_1d [mV]   |    bias_2a [mV]   |    bias_2b [mV]   |    bias_2c [mV]   |
|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|bbbb bbbb|bbbb bbbb|cccc cccc|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 56 | Byte 57 | Byte 58 | Byte 59 | Byte 60 | Byte 61 | Byte 62 | Byte 63 |

|    bias_2d [mV]   |ErrorCode|ErrPrm.LW|ErrPrm.HW| S.Flags |Sys Stat.| CHKSUM  |
|dddd dddd|dddd dddd|pppp pppp|pppp pppp|pppp pppp|ffff ffff|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 64 | Byte 65 | Byte 66 | Byte 67 | Byte 68 | Byte 69 | Byte 70 | Byte 71 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x2F
Byte 2 – Length of the message – 72 bytes
Byte 3 – Mode of the Amplifier
- 0x1X – Amplifier Reset
- 0x2X – Init Mode
- 0x3X – Debug Mode
- 0x4X – Service Mode
- 0x5X – STB Mode
- 0x6X – OPR/RX Mode
- 0x7X – OPR/TX Mode
- 0x8X – ATAC (ATU Tune Antenna Change) Procedure
- 0x9X – Set System Parameters Menu
- 0xAX – Turn OFF Procedure
- 0xA2 – Extra Cooling State

Byte 4 – Amplifier Flags 1 (Low Byte)
- bit 7 – FLAG_BIAS_CTRL — Request for Bias Voltages ON/OFF Procedures
- bit 6 – FLAG_PwrOff_Request — Final Request for Power OFF
- bit 5 – FLAG_Soft_PwrOff — Request for Power OFF by Software
- bit 4 – FLAG_PWRON_RS — Amplifier is Turned ON by RS232
- bit 3 – FLAG_PWRON_RMT — Amplifier is Turned ON by Remote Voltage
- bit 2 – FLAG_PWRON_BTN — Amplifier is Turned ON by Panel Button
- bit 1 – FLAG_PSU2_TEMPHI — PSU2 Over-Heating
- bit 0 – FLAG_PSU1_TEMPHI — PSU1 Over-Heating

Byte 5 – Amplifier Flags 1 (High Byte)
- bit 15 – FLAG_Last_Com — Last Command Source (0 - Local / 1 - Remote)
- bit 14 – FLAG_ATAC_InProgress — ATAC Procedure is in Progress
- bit 13 – FLAG_ATAC_REQ — Request for ATAC Procedure
- bit 12 – FLAG_HV_ON — Status of High Voltage Control Signal
- bit 11 – FLAG_HV_MON — Enable/Disable High Voltage Monitoring
- bit 10 – FLAG_HV_CTRL — Request for High Voltage ON/OFF Procedures
- bit 9 – FLAG_BIASEN — Status of Bias Control Signal
- bit 8 – FLAG_BIAS_MON — Enable/Disable Bias Voltages Monitoring

Byte 6 – Amplifier Flags 2 (Low Byte)
- bit 7 – FLAG_ORC_MON — Enable/Disable ORC Monitoring
- bit 6 – FLAG_OUTR — Status of the Output Relay Control Signal
- bit 5 – FLAG_OUTR_CTRL — Request for Output Relay ON/OFF Procedures
- bit 4 – FLAG_TX_ACCESS — Disable/Enable “OPR/TX Mode” Access

bit 3 – FLAG_INR Status of the Input Relay Control Signal
bit 2 – FLAG_INR_CTRL Request for Input Relay ON/OFF Procedures
bit 1 – FLAG_KEYIN Detected Level of the KEY IN Signal
bit 0 – FLAG_KEYIN_MON Enable/Disable KEY IN Monitoring

Byte 7 – Amplifier Flags 2 (High Byte)
bit 15 – FLAG_WRN_DIS Disable Warning Messages
bit 14 – FLAG_SF_ERR_DIS Disable SOFT FAULT Errors
bit 13 – FLAG_HF_ERR_DIS Disable HARD FAULT Errors
bit 12 – FLAG_I_LKG_MON ON/OFF Alarms Check During Input Relay Switching
bit 11 – FLAG_ASEL_TUN Indicate Whether ASEL is Tuned or Not (if associated)
bit 10 – FLAG_ATU_TUN Indicate Whether ATU is Tuned or Not (if associated)
bit 9 – FLAG_LPF_TUN Indicate Whether LPFs are Tuned or Not
bit 8 – FLAG_ORC ORC Condition After ORC Detection Procedure

Byte 8 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 9 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 10 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 11 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 12 – System Clock – High Word [sec] (Low Byte)
Byte 13 – System Clock – High Word [sec] (High Byte)
Byte 14 – System Clock – Low Word [sec] (Low Byte)
Byte 15 – System Clock – Low Word [sec] (High Byte)
Byte 16 – Temperature of PAM1 [deg K] (Low Byte)
Byte 17 – Temperature of PAM1 [deg K] (High Byte)
Byte 18 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 19 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 20 – Input Power [10 x W] (Low Byte)
Byte 21 – Input Power [10 x W] (High Byte)
Byte 22 – Forward Power [W] (Low Byte)
Byte 23 – Forward Power [W] (High Byte)
Byte 24 – Reflected Power [W] (Low Byte)
Byte 25 – Reflected Power [W] (High Byte)
Byte 26 – SWR [100x] (Low Byte)
Byte 27 – SWR [100x] (High Byte)
Byte 28 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte)
Byte 29 – PAM1 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte)
Byte 30 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (Low Byte) (Not Used)
Byte 31 – PAM2 Dissipation Power = P_OUT – P_DC [10 x W] (High Byte) (Not Used)
Byte 32 – bdata [mV] (Low Byte)
Byte 33 – bdata [mV] (High Byte)
Byte 34 – disbalance [mV] (Low Byte) (Not Used)
Byte 35 – disbalance [mV] (High Byte) (Not Used)
Byte 36 – VCC5 [mV] (Low Byte)
Byte 37 – VCC5 [mV] (High Byte)
Byte 38 – VCC26 [10 x V] (Low Byte)
Byte 39 – VCC26 [10 x V] (High Byte)
Byte 40 – HV1 [10 x V] (Low Byte)
Byte 41 – HV1 [10 x V] (High Byte)
Byte 42 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 43 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 44 – ID1 [mA] (Low Byte)
Byte 45 – ID1 [mA] (High Byte)
Byte 46 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 47 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 48 – Carrier frequency [kHz] (Low Byte)

Byte 49 – Carrier frequency [kHz] (High Byte)
Byte 50 – bias_1a [mV] (Low Byte) (A600S)
Byte 51 – bias_1a [mV] (High Byte) (A600S)
Byte 52 – bias_1b [mV] (Low Byte) (A600S)
Byte 53 – bias_1b [mV] (High Byte) (A600S)
Byte 54 – bias_1c [mV] (Low Byte) (Not Used)
Byte 55 – bias_1c [mV] (High Byte) (Not Used)
Byte 56 – bias_1d [mV] (Low Byte) (Not Used)
Byte 57 – bias_1d [mV] (High Byte) (Not Used)
Byte 58 – bias_2a [mV] (Low Byte) (Not Used)
Byte 59 – bias_2a [mV] (High Byte) (Not Used)
Byte 60 – bias_2b [mV] (Low Byte) (Not Used)
Byte 61 – bias_2b [mV] (High Byte) (Not Used)
Byte 62 – bias_2c [mV] (Low Byte) (Not Used)
Byte 63 – bias_2c [mV] (High Byte) (Not Used)
Byte 64 – bias_2d [mV] (Low Byte) (Not Used)
Byte 65 – bias_2d [mV] (High Byte) (Not Used)
Byte 66 – Error code of the alarm condition displayed on the amplifier's screen
Byte 67 – Parameter of the alarm condition (Low Byte)
Byte 68 – Parameter of the alarm condition (High Byte)
Byte 69 – Status flags for Fan speed and active LPF channel
bit 4 – bit 7 FAN Speed [0..4]
bit 0 – bit 3 Active FLP channel (low pass filter) [0..10]
Byte 70 – System status flags
bit 7 –
bit 6 –
bit 5 –
bit 4 –
bit 3 –
bit 2 –
bit 1 –
bit 0 – Error Code is for last Hard Fault Error (User_Flags, bit 15 is active)

Byte 71 – CHKSUM of the message

4. HARD FAULT event record

When Hard Fault Event occurred, list of parameters of the amplifier at this moment are saved in non volatile EEPROM memory in the main microcontroller of the amplifier. Information about Hard Fault event record are represents with four messages. Each of them is generated once on request!

4.1. Hard Fault Error Event - Record 1

Start TAG	Address	Length	HF Count	HF NUM	Error Codes 0	
0101 0101	0011 0001	0010 0110	cccc cccc	nnnn nnnn	eeee eeee	eeee eeee
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

Error Codes 1		Error Codes 2		Error Codes 3		Error Codes 4	
eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee	eeee eeee
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

```
|   Error Codes 5   |   Error Codes 6   |   Error Codes 7   |   Error Codes 8   |
|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 |

|   Error Codes 9   |   X   SPARE   X   |  Amplifier State  |   Amp Jump State  |
|eeee eeee|eeee eeee|ssss ssss|ssss ssss|ssss ssss|ssss ssss|jjjj jjjj|jjjj jjjj|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 23 | Byte 24 | Byte 25 | Byte 26 | Byte 27 | Byte 28 | Byte 29 | Byte 30 |

|     Clock HW      |     Clock LW      |    HF1 CHKSUM     | CHKSUM  |
|hhhh hhhh|hhhh hhhh|llll llll|llll llll|hhhh hhhh|hhhh hhhh|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 31 | Byte 32 | Byte 33 | Byte 34 | Byte 35 | Byte 36 | Byte 37 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x31
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault Errors until now
Byte 4 – Number of Hard Fault Error in the EEPROM memory
Byte 5 – Error Codes 0 (Low Byte)
Byte 6 – Error Codes 0 (High Byte)
Byte 7 – Error Codes 1 (Low Byte)
Byte 8 – Error Codes 1 (High Byte)
Byte 9 – Error Codes 2 (Low Byte)
Byte 10 – Error Codes 2 (High Byte)
Byte 11 – Error Codes 3 (Low Byte)
Byte 12 – Error Codes 3 (High Byte)
Byte 13 – Error Codes 4 (Low Byte)
Byte 14 – Error Codes 4 (High Byte)
Byte 15 – Error Codes 5 (Low Byte)
Byte 16 – Error Codes 5 (High Byte)
Byte 17 – Error Codes 6 (Low Byte)
Byte 18 – Error Codes 6 (High Byte)
Byte 19 – Error Codes 7 (Low Byte)
Byte 20 – Error Codes 7 (High Byte)
Byte 21 – Error Codes 8 (Low Byte)
Byte 22 – Error Codes 8 (High Byte)
Byte 23 – Error Codes 9 (Low Byte)
Byte 24 – Error Codes 9 (High Byte)
Byte 25 – Spare
Byte 26 – Spare
Byte 27 – Amplifier mode (Low Byte)
Byte 28 – Amplifier mode (High Byte)
Byte 29 – Amplifier transition state (Low Byte)
Byte 30 – Amplifier transition state (High Byte)
Byte 31 – System counter at the Hard Fault event (High Word / Low Byte)
Byte 32 – System counter at the Hard Fault event (High Word / High Byte)
Byte 33 – System counter at the Hard Fault event (Low Word / Low Byte)
Byte 34 – System counter at the Hard Fault event (Low Word / High Byte)
Byte 35 –Check Sum of the Hard Fault Record 1 (Low Byte)
Byte 36 –Check Sum of the Hard Fault Record 1 (High Byte)
Byte 37 – CHKSUM of the message

4.2. Hard Fault Error Event – Record 2

Start TAG	Address	Length	HF Count	HF NUM	IN FREQ [kHz]	
0101 0101	0011 0010	0010 0110	cccc cccc	nnnn nnnn	iiii iiii	iiii iiii
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

Fwd Power [W]		Refl Power [W]		In Power [10 x W]		Disbalance [mV]	
ffff ffff	ffff ffff	rrrr rrrr	rrrr rrrr	iiii iiii	iiii iiii	dddd dddd	dddd dddd
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

VCC26 [10 x V]		VCC5 [mV]		HV1 [10 x V]		HV2 [10 x V]	
VVVV VVVV	VVVV VVVV	vvvv vvvv	vvvv vvvv	hhhh hhhh	hhhh hhhh	hhhh hhhh	hhhh hhhh
Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22

ID1 [mA]		ID2 [mA]		TEMP1 [deg K]		TEMP2 [deg K]	
IIII IIII	IIII IIII	iiii iiii	iiii iiii	TTTT TTTT	TTTT TTTT	tttt tttt	tttt tttt
Byte 23	Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30

P_DC1 [10 x W]		P_DC2 [10 x W]		HF2 CHKSUM		CHKSUM	
DDDD DDDD	DDDD DDDD	dddd dddd	dddd dddd	hhhh hhhh	hhhh hhhh	cccc cccc	
Byte 31	Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x32
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault errors until now
Byte 4 – Number of Hard Fault error in the EEPROM memory
Byte 5 – Carrier frequency [kHz] (Low Byte)
Byte 6 – Carrier frequency [kHz] (High Byte)
Byte 7 – Forward Power [W] (Low Byte)
Byte 8 – Forward Power [W] (High Byte)
Byte 9 – Reflected Power [W] (Low Byte)
Byte 10 – Reflected Power [W] (High Byte)
Byte 11 – Input Power [10 x W] (Low Byte)
Byte 12 – Input Power [10 x W] (High Byte)
Byte 13 – Disbalance [mV] (Low Byte) (Not Used)
Byte 14 – Disbalance [mV] (High Byte) (Not Used)
Byte 15 – VCC26 [10 x V] (Low Byte)
Byte 16 – VCC26 [10 x V] (High Byte)
Byte 17 – VCC5 [mV] (Low Byte)
Byte 18 – VCC5 [mV] (High Byte)
Byte 19 – HV1 [10 x V] (Low Byte)
Byte 20 – HV1 [10 x V] (High Byte)
Byte 21 – HV2, only if PAM2 Exsist [10 x V] (Low Byte) (Not Used)
Byte 22 – HV2, only if PAM2 Exsist [10 x V] (High Byte) (Not Used)
Byte 23 – ID1 [mA] (Low Byte)
Byte 24 – ID1 [mA] (High Byte)
Byte 25 – ID2, only if PAM2 Exsist [mA] (Low Byte) (Not Used)
Byte 26 – ID2, only if PAM2 Exsist [mA] (High Byte) (Not Used)
Byte 27 – Temperature of PAM1 [deg K] (Low Byte)
Byte 28 – Temperature of PAM1 [deg K] (High Byte)

Byte 29 – Temperature of PAM2 [deg K] (Low Byte) (Not Used)
Byte 30 – Temperature of PAM2 [deg K] (High Byte) (Not Used)
Byte 31 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (Low Byte)
Byte 32 – P_DC1 - DC Power Consumption by PAM1 [10 x W] (High Byte)
Byte 33 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (Low Byte) (Not Used)
Byte 34 – P_DC2 - DC Power Consumption by PAM2 [10 x W] (High Byte) (Not Used)
Byte 35 – Check Sum of the Hard Fault Record 2 (Low Byte)
Byte 36 – Check Sum of the Hard Fault Record 2 (High Byte)
Byte 37 – CHKSUM of the message

4.3. Hard Fault Error Event – Record 3

Start TAG	Address	Length	HF Count	HF NUM	BIAS_1A [mV]	
0101 0101	0011 0011	0010 0110	cccc cccc	nnnn nnnn	aaaa aaaa	aaaa aaaa
Byte 0	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6

BIAS_1B [mV]		BIAS_1C [mV]		BIAS_1D [mV]		BIAS_2A [mV]	
bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc	dddd dddd	dddd dddd	aaaa aaaa	aaaa aaaa
Byte 7	Byte 8	Byte 9	Byte 10	Byte 11	Byte 12	Byte 13	Byte 14

BIAS_2B [mV]		BIAS_2C [mV]		BIAS_2D [mV]		P_DIS1 [10 x W]	
bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc	dddd dddd	dddd dddd	dddd dddd	dddd dddd
Byte 15	Byte 16	Byte 17	Byte 18	Byte 19	Byte 20	Byte 21	Byte 22

P_DIS2 [10 x W]		SWR [100x]		bdata [mV]		CAT Settings 1	
ssss ssss	ssss ssss	bbbb bbbb	bbbb bbbb	cccc cccc	cccc cccc	cccc cccc	cccc cccc
Byte 23	Byte 24	Byte 25	Byte 26	Byte 27	Byte 28	Byte 29	Byte 30

CAT Settings 2		CAT Settings 3		HF3 CHKSUM		CHKSUM	
cccc cccc	cccc cccc	cccc cccc	cccc cccc	hhhh hhhh	hhhh hhhh	cccc cccc	
Byte 31	Byte 32	Byte 33	Byte 34	Byte 35	Byte 36	Byte 37	

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x33
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault errors until now
Byte 4 – Number of Hard Fault error in the EEPROM memory
Byte 5 – bias_1a [mV] (Low Byte) (A600S)
Byte 6 – bias_1a [mV] (High Byte) (A600S)
Byte 7 – bias_1b [mV] (Low Byte) (A600S)
Byte 8 – bias_1b [mV] (High Byte) (A600S)
Byte 9 – bias_1c [mV] (Low Byte) (Not Used)
Byte 10 – bias_1c [mV] (High Byte) (Not Used)
Byte 11– bias_1d [mV] (Low Byte) (Not Used)
Byte 12 – bias_1d [mV] (High Byte) (Not Used)
Byte 13 – bias_2a [mV] (Low Byte) (Not Used)
Byte 14 – bias_2a [mV] (High Byte) (Not Used)
Byte 15 – bias_2b [mV] (Low Byte) (Not Used)
Byte 16 – bias_2b [mV] (High Byte) (Not Used)
Byte 17 – bias_2c [mV] (Low Byte) (Not Used)

Byte 18 – bias_2c [mV] (High Byte) (Not Used)
Byte 19 – bias_2d [mV] (Low Byte) (Not Used)
Byte 20 – bias_2d [mV] (High Byte) (Not Used)
Byte 21 – P_DIS1 [10 x W] (Low Byte) (A600S)
Byte 22 – P_DIS1 [10 x W] (High Byte) (A600S)
Byte 23 – P_DIS2 [10 x W] (Low Byte) (Not Used)
Byte 24 – P_DIS2 [10 x W] (High Byte) (Not Used)
Byte 25 – SWR [100x] (Low Byte)
Byte 26 – SWR [100x] (High Byte)
Byte 27 – Analog Band Data [mV] (Low Byte)
Byte 28 – Analog Band Data [mV] (High Byte)
Byte 29 – CAT Settings 1 (Low Byte) –
bit 4..bit 7 – CAT baud rate (only for CAT interface with TTL Levels or RS232 Levels)
1 – 1200 bps
2 – 4800 bps
3 – 9600 bps
4 – 19200 bps
5 – 38400 bps
6 – 57600 bps
bit0..bit3 – Not Used
Byte 30 – CAT Settings 1 (High Byte)
bit 4..bit 7 – CAT Interface Type
0 – OFF
1 – RS232 Levels
2 – TTL Levels
3 – BCD Code
4 – RS232 Levels
bit0..bit3 – CAT Command Set
0 – CAT interface not specified
1 – Set 1
...
15 – Set 15
Byte 31 – CAT Settings 2 (Low Byte)
Byte 32 – CAT Settings 2 (High Byte) – Byte Spacing [us] (0..65535)
Byte 33 – CAT Settings 3 (Low Byte)
Byte 34 – CAT Settings 3 (High Byte) – Polling Time [ms] (0..65535)
Byte 35 – Check Sum of the Hard Fault Record 3 (Low Byte)
Byte 36 – Check Sum of the Hard Fault Record 3 (High Byte)
Byte 37 – CHKSUM of the message

4.4. Hard Fault Error Event – Record 4

```
|Start TAG| Address | Length  |HF Count | HF NUM  | BIAS_1A_SET [mV]  |
|0101 0101|0011 0100|0010 0110|cccc cccc|nnnn nnnn|aaaa aaaa|aaaa aaaa|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  |

| BIAS_1B_SET [mV]  | BIAS_1C_SET [mV]  | BIAS_1D_SET [mV]  | BIAS_2A_SET [mV]  |
|bbbb bbbb|bbbb bbbb|cccc cccc|cccc cccc|dddd dddd|dddd dddd|aaaa aaaa|aaaa aaaa|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 7  | Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 |

| BIAS_2B_SET [mV]  | BIAS_2C_SET [mV]  | BIAS_2D_SET [mV]  |   Error Source    |
|bbbb bbbb|bbbb bbbb|cccc cccc|cccc cccc|dddd dddd|dddd dddd|eeee eeee|eeee eeee|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
```

```
| Byte 15 | Byte 16 | Byte 17 | Byte 18 | Byte 19 | Byte 20 | Byte 21 | Byte 22 |

|   X   SPARE   X   |     Aset|Atyp Anum|    User Flags     |     LPF State     |
|xxxx xxxx|xxxx xxxx|ssss ssss|tttt nnnn|ffff ffff|ffff ffff|llll llll|llll llll|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 23 | Byte 24 | Byte 25 | Byte 26 | Byte 27 | Byte 28 | Byte 29 | Byte 30 |

| Amplifier Flags 1 | Amplifier Flags 2 |    HF4 CHKSUM     | CHKSUM  |
|ffff ffff|ffff ffff|ffff ffff|ffff ffff|hhhh hhhh|hhhh hhhh|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 31 | Byte 32 | Byte 33 | Byte 34 | Byte 35 | Byte 36 | Byte 37 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x34
Byte 2 – Length of the message – 38 bytes
Byte 3 – Number of all Hard Fault errors until now
Byte 4 – Number of Hard Fault error in the EEPROM memory
Byte 5 – BIAS_1A_SET [mV] (Low Byte) (Not Used)
Byte 6 – BIAS_1A_SET [mV] (High Byte) (Not Used)
Byte 7 – BIAS_1B_SET [mV] (Low Byte) (Not Used)
Byte 8 – BIAS_1B_SET [mV] (High Byte) (Not Used)
Byte 9 – BIAS_1C_SET [mV] (Low Byte) (Not Used)
Byte 10 – BIAS_1C_SET [mV] (High Byte) (Not Used)
Byte 11 – BIAS_1D_SET [mV] (Low Byte) (Not Used)
Byte 12 – BIAS_1D_SET [mV] (High Byte) (Not Used)
Byte 13 – BIAS_2A_SET [mV] (Low Byte) (Not Used)
Byte 14 – BIAS_2A_SET [mV] (Hugh Byte) (Not Used)
Byte 15 – BIAS_2B_SET [mV] (Low Byte) (Not Used)
Byte 16 – BIAS_2B_SET [mV] (High Byte) (Not Used)
Byte 17 – BIAS_2C_SET [mV] (Low Byte) (Not Used)
Byte 18 – BIAS_2C_SET [mV] (High Byte) (Not Used)
Byte 19 – BIAS_2D_SET [mV] (Low Byte) (Not Used)
Byte 20 – BIAS_2D_SET [mV] (High Byte) (Not Used)
Byte 21 – Error Source (Low Byte)
Byte 22 – Error Source (High Byte)
Byte 23 – SPARE
Byte 24 – SPARE
Byte 25 – ATU Settings (Low Byte). Valid only when ATU is Associated
Byte 26 – ATU Settings (High Byte). Valid only when ATU is Associated
Byte 27 – User Flags (Low Byte):
- bit 7 – (0x07)
- bit 6 – (0x06) Operate Access 0 – Disabled 1 – Enabled
- bit 5 – (0x05) ON_BTN Pulses 0 – Long ON / Short OFF 1 – Short ON / Long OFF
- bit 4 – (0x04) ATAC Procedure: 0 – Manual 1 – Auto
- bit 3 – (0x03) CPU Buzzer: 0 – Disabled 1 – Enabled
- bit 2 – (0x02) Temperature Flag: 0 – [º C] 1 – [º F]
- bit 1 – (0x01) Auto Operate: 0 – Disabled 1 – Enabled
- bit 0 – (0x00) Extra Cooling: 0 – Disabled 1 – Enabled

Byte 28 – User Flags (High Byte):
- bit 7 – (0x0F) Last Power OFF is because of HARD FAULT Error
- bit 6 – (0x0E)
- bit 5 – (0x0D)
- bit 4 – (0x0C)
- bit 3 – (0x0B)
- bit 2 – (0x0A)

bit 1 – (0x09)
bit 0 – (0x08)
Byte 29 – State of the Low Pass Filters (Low Byte)
Byte 30 – State of the Low Pass Filters (High Byte)
Byte 31 – Amplifier Flags
Byte 32 – Amplifier Flags
Byte 33 – Amplifier Flags
Byte 34 – Amplifier Flags
Byte 35 – Check Sum of the Hard Fault Record 4 (Low Byte)
Byte 36 – Check Sum of the Hard Fault Record 4 (High Byte)
Byte 37 – CHKSUM of the message

5. Messages and commands from the computer to the main CPU of the amplifier

5.1. Commands for amplifier management:

```
|Start TAG| Address | Length  | Amp CMD |Param. HW|Param. LW| Buttons | CHKSUM  |
|0101 0101|1000 0001|0000 1000|cccc cccc|pppp pppp|pppp pppp|bbbb bbbb|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Bute 7  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x81
Byte 2 – Length of the message – 8 bytes
Byte 3 – Command to the amplifier
- 0x00 – Empty command
- 0x01 – Request for specific message
  - Byte 4 – Number of desired message
  - Byte 5 – Number of desired message
- 0x02 – Request for amplifier mode change
  - Byte 4 – No matter
  - Byte 5 – Number of desired amplifier mode:
    - 0x01 – Amplifier restart
    - 0x02 – Amplifier restart
    - 0x03 – Debug Mode
    - 0x04 – Service Mode
    - 0x05 – STB Mode
    - 0x06 – OPR/RX Mode
    - 0x07 – OPR/TX Mode
    - 0x08 – “Antenna Tune / Antenna Change” Procedure (ATAC)
    - 0x09 – Menu “Set System Parameters”
    - 0x0A – Turn OFF the Amplifier
    - 0x40 – Prohibit OPR/TX mode
    - 0x80 – Allow OPR/TX mode
- 0x03 – Request for specific Hard Fault Event Record
  - Byte 4 – No matter
  - Byte 5 – Number of HARD FAULT event record
- 0x04 – Update user settings: User Flags
  - Byte 4 – User_Flags (HW)
  - Byte 5 – User_Flags (LW)
    - bit 15 - Last Power OFF is because of HARD FAULT Error (system flag)
    - bit 14 -

bit 13 -
bit 12 -
bit 11 -
bit 10 -
bit 9 -
bit 8 -
bit 7 -

bit 6 - Operate Access	0 – Disabled	1 – Enabled
bit 5 - ON_BTN Pulses	0 - Long ON / Short OFF	1 - Short ON / Long OFF
bit 4 - ATAC Procedure	0 - Manual	1 - Auto
bit 3 - CPU Buzzer	0 - Disabled	1 - Enabled
bit 2 - Temperaure Flag	0 - [deg C]	1 - [deg F]
bit 1 - Auto Operate	0 - Disabled	1 - Enabled
bit 0 - Extra Cooling	0 - Disabled	1 - Enabled

0x05 –CAT interface setting: CAT Settings 1:
Byte 4 – CAT Settings 1 (HW)
Byte 5 – CAT Settings 1 (LW)
bits [15..12] - CAT Interface Type:
0 - None
1 - RS232
2 - TTL
3 - BCD
4 - Analog
bits [11..8] – CAT Command Set:
0 - None
1 - ICOM
2 - YEASU FT-450, ...
3 - YEASU FT-817, FT-897, ...
4 - YEASU FT-1000MP, ...
5 - ELECRAFT K3, KENWOOD TS-480 ...
6 - ...
bits [7..4] - CAT_Baud_Rate:
0 - None
1- 1200
2 - 4800
3 - 9600
4 - 19200
5 - 38400
6 - 57600
0x06 –CAT interface setting: CAT Settings 2: Byte Spacing [us]
Byte 4 – CAT Settings 2 (HW)
Byte 5 – CAT Settings 2 (LW)
0x07 –CAT interface setting: CAT Settings 3: Polling Time [ms]
Byte 4 – CAT Settings 3 (HW)
Byte 5 – CAT Settings 3 (LW)
0x08 – Clear SOFT FAULT errors and flag for last Power OFF is because of HARD FAULT Error (User Flags, bit 15)
Byte 4 – No matter
Byte 5 – No matter
0x09 – Command for manual changing of antenna and band
Byte 4 – Number of antenna
Byte 5 – Number of band:
0x01 – 0x0A – Number of desired band
0x80 – Command “Next Band”

0x40 – Command „Previous Band”

0x0A – Command for Buzzer volume and Buzzer sound signal generation

Byte 4 – Sound signal choice

0 – No signal

1 – Signal	BUZZER_PULSE	(Single Pulse)
2 – Signal	BUZZER_WRN	(Warning signal)
3 – Signal	BUZZER_PWROFF	(Amplifier Power OFF signal)
4 – Signal	BUZZER_SF	(Signal for SOFT FAULT error)
5 – Signal	BUZZER_HF	(Signal for HARD FAULT error)
6 – Signal	BUZZER_PWRON	(Amplifier Power ON signal)
ASCII –	BUZZER_LETTER	(Morze code coresponding to the ASCII)

Byte 5 –Buzzer volume (0 – 100 %)

0x0B – Command for entire LOG file to be sent trough RS232 interface of the amplifier

Byte 4 – No matter

Byte 5 – No matter

0x0C – Commands when amplifier is in “Service Mode”

Byte 4 – Select module test

Byte 5 – Module test control

Byte 4		**Byte 5**		**Note**
0x00	Invalid value	0x00	Invalid Value	
0x01	Idle Current	0x00 0x01 0xFF	OFF ON Deactivate all tests	
0x02	High Voltage	0x00 0x01 0xFF	OFF ON Deactivate all tests	
0x03	Relay Voltage	0xFF	Deactivate all tests	Not Used
0x04	Band Select Relays Tests	0x00 0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08 0x09 0x0A 0x80 0x40 0xFF	OFF 160m 80m 40m 30m 20m 17m 15m 12m 10m 6m Command: Next Command: Previous Deactivate all tests	 United with 15m United with 17m United with 10m United with 12m
0x05	Input Relay Test	0x00 0x01	OFF ON	CPU Board returns the result
0x06	Output Relay Test	0x00 0x01	OFF ON	CPU Board returns the result

0x07	FAN Speed	0x00 0x01 0x02 0x03 0x80 0x40 0xFF	OFF Speed 1 Speed 2 Speed 3 Command: Next Command: Previous Deactivate all tests	If the temperature is high, command for lower fan speeds might be ignored.
0xFF	Deactivate all tests	X		

0x0D – Commands for default factory settings restore

Byte 4 – No matter

Byte 5 – Select the parameters to be restored

1 – Reset User Preferences to Factory Defaults

2 – Reset Working Hours Counter

3 – Erase Errors Log

4 – Reset All to Factory Defaults

Byte 4 – The parameter depends of the command - Byte 3. (HIGH WORD)
Byte 5 – The parameter depends of the command - Byte 3. (LOW WORD)
Byte 6 – Amplifier's keyboard state

bit 7 –
bit 6 –
bit 5 – Button 6 state
bit 4 – Button 5 state
bit 3 – Button 4 state
bit 2 – Button 3 state
bit 1 – Button 2 state
bit 0 – Button 1 state

Byte 7 – CHKSUM of the message

5.2. Serial number of the amplifier

```
|Start TAG| Address | Length  |A.SN [0] |A.SN [1] |A.SN [2] |A.SN [3] |A.SN [4] |
|0101 0101|1000 0010|0001 0000|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|A.SN [5] |A.SN [6] |A.SN [7] |A.SN [8] |A.SN [9] |A.SN [10]|A.SN [11]| CHKSUM  |
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x82
Byte 2 – Length of the message – 16 bytes
Byte 3..Byte 14 – Serial number of the amplifier
Byte 15 – CHKSUM of the message

5.3. Hello message of the amplifier.

```
|Start TAG| Address | Length  |H.Msg[0] |H.Msg[1] |H.Msg[2] |H.Msg[3] |H.Msg[4] |
|0101 0101|1000 0011|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
```

```
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[5] |H.Msg[6] |H.Msg[7] |H.Msg[8] |H.Msg[9] |H.Msg[10]|H.Msg[11]|H.Msg[12]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[13]|H.Msg[14]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x83
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello message – chars from 0 to 14
Byte 18 – CHKSUM of the message

```
|Start TAG| Address | Length  |H.Msg[15]|H.Msg[16]|H.Msg[17]|H.Msg[18]|H.Msg[19]|
|0101 0101|1000 0100|0001 0011|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|H.Msg[20]|H.Msg[21]|H.Msg[22]|H.Msg[23]|H.Msg[24]|H.Msg[25]|H.Msg[26]|H.Msg[27]|
|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|ssss ssss|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|H.Msg[28]|H.Msg[29]| CHKSUM  |
|ssss ssss|ssss ssss|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x84
Byte 2 – Length of the message – 19 bytes
Byte 3 – Byte 17 – Hello message – chars from 15 to 29
Byte 18 – CHKSUM of the message

5.4. Antenna names

```
|Start TAG| Address | Length  |AntN AntT|Name [0] |Name [1] |Name [2] |Name [3] |
|0101 0101|1000 0101|0001 0011|nnnn tttt|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  | Byte 5  | Byte 6  | Byte 7  |

|Name [4] |Name [5] |Name [6] |Name [7] |Name [8] |Name [9] |Name [10]|Name [11]|
|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|nnnn nnnn|
|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 8  | Byte 9  | Byte 10 | Byte 11 | Byte 12 | Byte 13 | Byte 14 | Byte 15 |

|Name [12]|Name [13]| CHKSUM  |
|nnnn nnnn|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|
| Byte 16 | Byte 17 | Byte 18 |
```

Byte 0 – Start byte of the message – 0x55

Byte 1 – Address of the message – 0x85
Byte 2 – Length of the message – 19 bytes
Byte 3 – Number and type of the antenna
bit 0 – bit 3 – Type of the antenna [1 - Default, 2 - Exact, 3 - Narrow, 4 - Wide]
bit 4 – bit 7 – Number of the antenna [1..10]
Byte 4 ÷ Byte 17 – Name of the antenna
Byte 18 – CHKSUM of the message

5.5. Acknowledge after message receiving by main CPU module of the amplifier:

Generates automaticly after message receiving by main CPU module of the amplifier. If main CPU module don't receive this message as reply last message will be sent again.

```
|Start TAG| Address | Length  | Msg Num | CHKSUM  |
|0101 0101|1000 0110|0000 0101|nnnn nnnn|cccc cccc|
|---- ----|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  | Byte 4  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x86
Byte 2 – Length of the message – 5 bytes
Byte 3 – Number of received message
Byte 4 – CHKSUM of the message

5.6. Command for HEX messages format to ASCII messages format switching:

Note: Obsolete! (Removed)

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0000|0000 0100|0001 0111|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x90
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x17

Note: From ASCII message format to HEX messages format is switches with sign '~'

5.7. Disable Automatic telemetry data through RS232 interface of the amplifier:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0001|0000 0100|0001 0110|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x91
Byte 2 – Length ofteh amplifier – 4 bytes
Byte 3 – CHKSUM of the message – 0x16

5.8. Enable automatic telemetry data through RS232 interface of the amplifier:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0010|0000 0100|0001 0101|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x92
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x15

5.9. Bootloader activation:

```
|Start TAG| Address | Length  | CHKSUM  |
|0101 0101|1001 0011|0000 0100|0001 0100|
|---- ----|---- ----|---- ----|---- ----|
| Byte 0  | Byte 1  | Byte 2  | Byte 3  |
```

Byte 0 – Start byte of the message – 0x55
Byte 1 – Address of the message – 0x93
Byte 2 – Length of the message – 4 bytes
Byte 3 – CHKSUM of the message – 0x14